Project Design Phase-II

Technology Stack (Architecture & Stack)

Team ID	6ab8c2c230f9767f87730236
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Technical Architecture

Three-tier architecture: User Interface → Application/API Layer → Database. The application logic evaluates field values and applies script-controlled ACL rules before allowing record read/update access. An audit component records access decisions.

S.No	Component	Description	Technology
1	User Interface	Web interface for administrators and authorized users to view records and configure ACL rules.	HTML, CSS, JavaScript, React.js
2	Application Logic-1	Authentication and authorization; validates user identity and role.	Python, Flask
3	Application Logic-2	Script-controlled ACL engine evaluates field values and access conditions.	Python rule engine / custom ACL scripts
4	Application Logic-3	REST API handles record requests and returns only permitted data.	Flask REST API
5	Database	Stores users, roles, records, field values and ACL rules.	MySQL
6	Cloud Database	Optional managed database deployment for production.	MySQL on cloud / managed SQL
7	File Storage	Stores configuration exports, logs or supporting documents when required.	Local filesystem / cloud object storage
8	External API-1	Email/OTP notifications for authentication and access events.	SMTP / Email API
9	External API-2	Optional identity provider integration for login.	OAuth 2.0 / Google Identity
10	Machine Learning Model	Not required; access decisions are rule/script based.	Not applicable
11	Infrastructure	Deploy the web/API application and database on a local server or cloud VM/container.	Docker, Linux, Nginx

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use open-source web and API frameworks for rapid development and maintainability.	React.js, Flask, Python
2	Security	Authentication, role-based access, field-level ACL evaluation, input validation, HTTPS and audit logging.	OAuth 2.0, JWT, HTTPS/TLS, bcrypt, OWASP practices
3	Scalable Architecture	Separate UI, API/ACL engine and database layers so components can scale independently.	3-tier architecture, REST API, Docker
4	Availability	Application can be deployed with restart policies and a replicated/managed database for improved availability.	Docker, Nginx, cloud managed DB
5	Performance	Evaluate ACL rules before returning records, use indexed fields and pagination to reduce response size.	MySQL indexes, caching, pagination

Architecture Flow: User → React Web UI → Flask REST API → Authentication/ACL Script Engine → MySQL Database → Authorized Record Response. Access decisions are logged for auditing.